Retail SQL Project

Author: Musrat Alam

ER Diagram (Simple Text-Based)

```
Customers --< Orders --< OrderDetails >-- Products
```

Table Schemas

Customers

Column	Data Type	Key
CustomerID	INT	PRIMARY KEY
Name	VARCHAR(50)	
Email	VARCHAR(50)	
Phone	VARCHAR(15)	
City	VARCHAR(30)	

Products

Column	Data Type	Key
ProductID	INT	PRIMARY KEY
ProductName	VARCHAR(50)	
Category	VARCHAR(30)	
Price	DECIMAL(10,2)	
Stock	INT	

Orders

Column	Data Type	Key
OrderID	INT	PRIMARY KEY

Column	Data Type	Key
CustomerID	INT	FOREIGN KEY
OrderDate	DATE	

OrderDetails

Column	Data Type	Key
OrderDetailID	INT	PRIMARY KEY
OrderID	INT	FOREIGN KEY
ProductID	INT	FOREIGN KEY
Quantity	INT	

Sample SQL Queries

Total Sales per Product

```sql
SELECT p.ProductName, SUM(od.Quantity * p.Price) AS TotalSales
FROM OrderDetails od
JOIN Products p ON od.ProductID = p.ProductID
GROUP BY p.ProductName;
```

Customer Order Summary

```sql
SELECT c.Name, COUNT(o.OrderID) AS TotalOrders, SUM(od.Quantity * p.Price) AS
TotalSpent
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
JOIN OrderDetails od ON o.OrderID = od.OrderID
JOIN Products p ON od.ProductID = p.ProductID
GROUP BY c.Name;
```

Products Low in Stock (less than 20)

```sql
SELECT ProductName, Stock
FROM Products
WHERE Stock < 20;
```

Recent Orders in January 2026

```
SELECT o.OrderID, c.Name, o.OrderDate
FROM Orders o
JOIN Customers c ON o.CustomerID = c.CustomerID
WHERE MONTH(o.OrderDate) = 1 AND YEAR(o.OrderDate) = 2026;
```

View for Easy Reporting

```
CREATE VIEW CustomerOrders AS
SELECT c.Name AS CustomerName, p.ProductName, od.Quantity, (od.Quantity *
p.Price) AS TotalAmount, o.OrderDate
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
JOIN OrderDetails od ON o.OrderID = od.OrderID
JOIN Products p ON od.ProductID = p.ProductID;
```

Sample Query Outputs

Total Sales per Product

ProductName	TotalSales
Laptop	45000.00
Smartphone	40000.00
Notebook	250.00
Pen	150.00

Products Low in Stock

ProductName	Stock
Laptop	10

Customer Order Summary

Name	TotalOrders	TotalSpent
Alice Khan	2	45250.00
Rahim Ali	1	40000.00

Project Summary for Resume

Developed a Retail Store Management System using SQL to manage customers, products, orders, and inventory. Implemented relational tables, data integrity constraints, and queries for reporting sales, stock, and customer analytics. Created views to improve efficiency.